

Wave Pulse Reflection: Fixed vs Free Ends

Wave Pulse Reflection (Free & Fixed Ends) · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://ophysics.com/w9.html>

Curriculum links: waves

Learning intentions

- I can describe how a wave pulse inverts when it reflects from a fixed end.
- I can describe how a wave pulse does not invert when it reflects from a free end.
- I can relate these boundary behaviours to reflection of waves at real boundaries between media.

Getting started

Open the simulation and select the fixed-end setting first. Send a single pulse down the string and watch what happens when it reaches the end and returns.

Tasks

1. With the fixed end selected, send a pulse and sketch its shape both before and after reflection. Is the reflected pulse inverted?
2. Switch to the free end and repeat. Sketch the pulse before and after reflection. Is it inverted this time?
3. Change the width and amplitude of the pulse and repeat both reflections. Does changing the pulse shape change whether it inverts?
4. For a fixed string length, measure the time between sending the pulse and seeing it return, for both end types. Should this time be different for a fixed vs a free end? Explain your reasoning.

Extension (optional)

Sound waves reflect off a hard wall very differently to how they behave at an open doorway. Relate this to the fixed-end and free-end reflections you observed.

Generated 10 August 2026 by STEM Simulation Hub — AI-generated worksheet, not reviewed by a human. Verify curriculum accuracy and classroom suitability before use.